3D laser scanning, as-built models and the engineering that follows

We scan the hull or structure, build the as-built 3D model, and run the structural and stability work on that same model. All in one small shop in Newport, RI.

UEI	CAGE	BUSINESS TYPE	NAICS
NWJRCE69AWH9	Pending	Small business · Since 2018	541330 · 541370 · 541380 · 541690 · 541715

What we do

- **3D laser scanning.** Hulls, decks, appendages and structure, scanned with our own FARO Focus X 30.
- **As-built 3D models.** Surface models and drawings in Rhino and SolidWorks, ready for engineering, repair or new parts.
- **Structural analysis on the scan.** FEA on the as-built geometry in Femap/Nastran and SolidWorks Simulation.
- **Stability and USCG compliance.** Hydrostatics and stability in GHS and Orca3D, inclining experiments, and stability and rig-loads reports addressed to the USCG (46 CFR Subchapter T).
- **Deliverables.** Registered point cloud in E57 and native FARO format, a FARO SCENE registration report, the as-built model in Rhino or SolidWorks, and an engineering report when analysis is ordered.

Why Wyspa

- Scan, model and analysis in one shop. No hand-off between a scan vendor and an engineering firm.
- Naval architects run the scan, so the model comes back ready for engineering.
- We come to the vessel. Recent scans were done on site in a working repair yard. We handled yard access and insurance paperwork.

101 of 191
projects where scanning fed the engineering

85 ft
aluminum patrol boat, scanned on site at a repair yard (2026)

Track record (commercial)

- **85 ft aluminum patrol boat** (2026). Scanned on site in a working repair yard; as-built 3D model of the outer surfaces in progress.
- **USCG Subchapter T conversions.** A historic 12 Metre (USCG approved) and a 12-passenger Pearson 35 (structure approved, rig review pending). Scans and stability on both; incline and rig loads on the Pearson.
- Other scans: a 114 ft motor yacht, an aircraft fuselage and wings, keels, rudders and rigs.

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